

Manuscript outline (reworked framing)

enceladus_DSM studies/keff_sintering/manuscript_outline.tex

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Framing. Sintering of ice by vapor transport, and the bulk conductivity that results, under conditions a laboratory can reach; Enceladus is the motivation and the closing implication. Every claim stays inside the simulated range: -40 to -5 °C, air at 1 atm, sealed pores, $50\text{ }\mu\text{m}$ grains, 30 days, two dimensions. Themes below are prompts, not text. Figures are in `ManuscriptFigures/`; tables are commands in `Tables/tables.tex`. Notes behind this outline: `MANUSCRIPT_PLAN.md`.

Front matter

- **Key points** (three). (i) For a given packing, temperature only rescales time. (ii) Conductivity follows specific surface area by one power law in well-connected packings; porosity sets the level. (iii) The model is the vapor route alone; what it does and does not reproduce of laboratory neck growth.
- **Abstract; Plain Language Summary.** Written last.

1 Introduction

- **P1. The problem.** Ice aggregates change their thermal and mechanical properties as grains sinter; property evolution has been measured in the laboratory but not tied to microstructure.
- **P2. Mechanisms.** Sintering without densification; vapor transport (sublimation and desublimation) and surface diffusion. Define “desublimation” here and reserve “deposit” for plume material.

Figure 1 Mechanisms on a grain pair (a); the same process in an aggregate (b). Cite at the end of P2.

- **P3. What laboratory work has shown.** Molaro et al. (2019): neck growth, mechanisms by temperature. Choukroun et al. (2020): strengthening of sealed ice-powder samples, activation energy 24 kJ mol^{-1} . Demmenie et al. (2025): neck growth at saturation.
- **P4. Why not natural snow.** Angular crystals at far higher porosity; overburden and densification. The analogues are unloaded ice-sphere and ice-powder samples.
- **P5. Motivation.** Plume deposits on Enceladus: what a lander or a thermal measurement would need to know.
- **P6. This work.** Phase-field model of vapor-transport sintering, periodic homogenization for conductivity, a porosity–temperature matrix; the three results; road map.

2 Methods

2.1 Phase-field model of sintering by vapor transport

- Fields (ϕ, T, ρ_v) and governing equations; what is and is not in the model (no surface diffusion, no densification).
- Material properties: cite **Table S1** (`\TableMaterialProperties`).

2.2 Interface kinetics and the Gibbs–Thomson condition

- α_c and β_{sub} (always quoted together), $d_0, \rho_{vs}^I(T)$. Values: $\alpha_c = 1.020 \times 10^{-3}$ in the aggregates, 0.1 in the Molaro pairs, 1.002×10^{-3} in the saturated pair.
- The thin-interface term $\Delta\beta$ in τ_{sub} , as in Kaempfer & Plapp (2009) and Moure & Fu (2024); what it costs (realised coefficient $\beta_{\text{sub}} + \Delta\beta$).
- The time scale τ_{sub} and the sintering age $\theta = t/\tau_{\text{sub}}$; state that ε is fixed across the study. *Open: state the clock as τ_{sub} at fixed ε , or as $\tau_R = \bar{R}^2\beta_{\text{sub}}/d_0$.*

2.3 Homogenization for the effective thermal conductivity

- Cell problem, correctors, tensor conductivity law in the interface.

Figure 2 The method on one packing: cell, interface and mesh, corrector, heat flux. After the cell problem is stated.

2.4 Numerical implementation

- Discretization, time stepping, sampling of $\mathbf{K}_{\text{eff}}$. Cite **Table S2** (`\TableKeffParametersFull`).
- Checks, each one sentence with a pointer to the supplement: domain size (Fig. S1), interface width (Fig. S2), time step and sampling (Fig. S3).

2.5 Aggregates: packings, boundary conditions, simulation matrix

- Drop-and-roll deposition with void filling; five packings per porosity.
- Periodic boundaries: scale separation, no wall physics to invent, a closed system driven by curvature alone.
- The matrix: five porosities $\times$ five temperatures $\times$ five packings (125 runs). The master simulation used for single-run figures.

3 Results

3.1 Two-grain sintering

Unsaturated pair (Molaro et al., 2019).

- Set-up; cite **Table S3** (\TableGrainPairParameters).
- With the vapor loss matched to the measured shrinkage (a calibration at -20°C), the model gives 41% of the measured neck growth.
- -5°C : the authors' own caveats on that data (Sect. 4.1, Appendix B).

Figure 3 Sections, neck width, grain shrinkage. After the set-up paragraph.

Saturated pair (conditions of Demmenie et al., 2025).

- Equal grains, vapor saturated over the grains' own curvature; the grain size holds.
- Fitted exponent 0.21 after relaxation (0.21–0.23 by window), against 0.26–0.33 measured and $1/3$ for attachment-limited vapor transport. *Open: interpretation. Parallel mechanisms add rates, so a missing surface-diffusion term cannot raise an exponent; candidates are the mixed attachment/diffusion regime, the large initial neck, viscous flow of the quasi-liquid layer at -3°C .*

Figure 4 Sections and neck width with the free and one-third fits. After the result is stated.

3.2 Aggregates

Set-up.

- One paragraph: the matrix and what is held fixed.

Table 1 \TableKeffParameters. After the sentence that states the matrix.

What sintering does to the aggregate.

- Necks form, small pores round and close, the ice fraction is conserved; orientation only, no analysis.

Figure 5 Gallery: five porosities at four instants.

Temperature sets the clock.

- Define θ in use; 30 days spans 43 to $1256 \tau_{\text{sub}}$.

Table 2 \TableKeffTemperature. After θ is introduced, before Figure 6.

- Five temperatures give one curve for k_{eff} and for SSA; for every packing k_{eff} at matched SSA agrees within 0.27% (Fig. S4 for all packings).
- SSA falls linearly in $\log \theta$ after the transient, about 0.13 per decade; state the range, do not extrapolate.

Figure 6 Snapshots of the master packing; k_{eff} against days and against θ ; SSA against θ .

Conductivity as a function of the microstructural state.

- Porosity dominates the level (a factor of about 3.3 across the range at 30 d); packing-to-packing scatter is comparable to the gap between neighbouring porosities.
- One power law in SSA, exponent -0.80 , for porosity ≤ 0.375 ; the level falls exponentially with porosity, 6.6% scatter.

- Above 0.40 the exponent weakens and scatters; no claim made there.
- Rise table by porosity and temperature (in text or Fig. S5); the two densest packings tie.

Figure 7 $k_{\text{eff}}(t)$ by porosity; k_{eff} against SSA; level; exponent.

4 Discussion

4.1 Temperature dependence of the kinetics

- Why the collapse holds: attachment-limited transfer, fixed α_c , one discretization; what would break it.
- Activation energy of the vapor route, about 50 kJ mol⁻¹, against 24 measured by Choukroun et al. for strengthening; the ranges overlap at 233–243 K, so this is a test an experiment can apply.

4.2 SSA as the state variable

- What the power law means and where it comes from; why porosity alone is not enough (packing scatter, connectivity).
- Loss of the law at high porosity: poorly connected ice, amplified in two dimensions. Anisotropy as a consistency check (one sentence).

4.3 What the vapor route captures and what it lacks

- Read Figures 3 and 4 together: part of the measured neck growth, a lower exponent at saturation.
- Surface diffusion and the quasi-liquid layer are absent; what each would change (rate versus exponent). Literature review needed before any claim.

4.4 What a laboratory could measure

- The three predictions as experiments: time–temperature superposition and its activation energy; k_{eff} against SSA on unloaded ice-sphere samples; SSA against log time.

4.5 Implications for plume deposits on Enceladus

- Qualitative only: sealed near-subsurface (about a metre down) as the closest setting; which conclusions carry (state dependence, exponent) and which do not (rates at 60–180 K, absolute conductivity).
- No timescales in years. Small-sensor point, one or two sentences.

4.6 Limitations

- Two dimensions; air-filled pores at 1 atm; one grain size; vapor route only; first day or so depends on the interface width; claims for porosity ≤ 0.375 .

5 Conclusions

- Three results, each with its range of validity; the single most useful experiment to do next.

Open Research; Acknowledgments

Code and data availability; run inventory (`RUN_TABLE.md`).

Supporting Information

- **Tables.** S1 material properties; S2 complete simulation parameters; S3 two-grain parameters.
- **Figures.** S1 domain-size convergence (and k_{xy} against size); S2 interface width; S3 time step and sampling; S4 temperature collapse for all packings; S5 porosity trends, SSA by porosity, anisotropy in time.
- **Text.** Packing construction; run table.